

Contact Hamiltonian systems: applications to thermodynamics

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PART I: Contact Hamiltonian systems

Different geometries provide different dynamics

Dynamics on symplectic geometry

As it is well known, Hamiltonian dynamics are developed using symplectic geometry. Indeed, let (M, ω) be a symplectic manifold, that is, ω is a non-degenerate closed 2-form, say $d\omega = 0$ and $\omega^n \neq 0$, where M has even dimension $2n$. Then, if $H : M \rightarrow \mathbb{R}$ is a Hamiltonian function, the Hamiltonian vector field X_H is obtained using the equation

$$\flat(X_H) = dH \quad (1)$$

where $\flat$ is the vector bundle isomorphism

$$\flat : TM \rightarrow T^*M, \quad \flat(v) = i_v \omega$$

In Darboux coordinates (q^i, p_i) we have $\omega = dq^i \wedge dp_i$ and

$$X_H = \frac{\partial H}{\partial p_i} \frac{\partial}{\partial q^i} - \frac{\partial H}{\partial q^i} \frac{\partial}{\partial p_i}$$

in such a way that an integral curve $(q^i(t), p_i(t))$ satisfies the Hamilton equations

$$\frac{dq^i}{dt} = \frac{\partial H}{\partial p_i}, \quad \frac{dp_i}{dt} = -\frac{\partial H}{\partial q^i} \quad (2)$$

Dynamics on cosymplectic geometry

A cosymplectic structure on an odd-dimensional manifold M is a pair (Ω, η) where Ω is a closed 2-form, η is a closed 1-form, and $\eta \wedge \Omega^n \neq 0$, where M has dimension $2n + 1$. (M, Ω, η) will be called a cosymplectic manifold.

There is a Darboux theorem for a cosymplectic manifold, that is, there are local coordinates (called Darboux coordinates) (q^i, p_i, z) around any point of M such that

$$\Omega = dq^i \wedge dp_i, \quad \eta = dz$$

There exists a unique vector field (called Reeb vector field) $\mathcal{R}$ such that

$$i_{\mathcal{R}} \Omega = 0, \quad i_{\mathcal{R}} \eta = 1$$

In Darboux coordinates we have

$$\mathcal{R} = \frac{\partial}{\partial z}$$

Let $H : M \rightarrow \mathbb{R}$ be a Hamiltonian function, say $H = H(q^i, p_i, z)$.

Consider the vector bundle isomorphism

$$\tilde{b} : TM \longrightarrow T^*M, \quad \tilde{b}(v) = i_v \Omega + \eta(v) \eta$$

and define the gradient of H by

$$\tilde{b}(\text{grad } H) = dH$$

Then

$$\text{grad } H = \frac{\partial H}{\partial p_i} \frac{\partial}{\partial q^i} - \frac{\partial H}{\partial q^i} \frac{\partial}{\partial p_i} + \frac{\partial H}{\partial z} \frac{\partial}{\partial z} \quad (3)$$

Next we can define two more vector fields:

- The Hamiltonian vector field

$$X_H = \text{grad } H - \mathcal{R}(H)\mathcal{R}$$

- and the evolution vector field

$$\mathcal{E}_H = X_H + \mathcal{R}$$

From (3) we obtain the local expression

$$\mathcal{E}_H = \frac{\partial H}{\partial p_i} \frac{\partial}{\partial q^i} - \frac{\partial H}{\partial q^i} \frac{\partial}{\partial p_i} + \frac{\partial}{\partial z} \quad (4)$$

Therefore, an integral curve $(q^i(t), p_i(t), z(t))$ of $\mathcal{E}_H$ satisfies the time-dependent Hamilton equations

$$\frac{dq^i}{dt} = \frac{\partial H}{\partial p_i} \quad (5)$$

$$\frac{dp_i}{dt} = -\frac{\partial H}{\partial q^i} \quad (6)$$

$$\frac{dz}{dt} = 1 \quad (7)$$

and then $z = t + \text{const}$ so that both coordinates can be identified.

Dynamics on contact geometry

Consider now a contact manifold (M, η) with contact form η ; this means that $\eta \wedge d\eta^n \neq 0$ and M has odd dimension $2n + 1$. There exists a unique vector field $\mathcal{R}$ (also called Reeb vector field) such that

$$i_{\mathcal{R}} d\eta = 0, \quad i_{\mathcal{R}} \eta = 1$$

There is a Darboux theorem for contact manifolds so that around each point in M one can find local coordinates (called Darboux coordinates) (q^i, p_i, z) such that

$$\eta = dz - p_i dq^i$$

In Darboux coordinates we have

$$\mathcal{R} = \frac{\partial}{\partial z}$$

Define now the vector bundle isomorphism

$$\bar{b} : TM \longrightarrow T^*M, \quad \bar{b}(v) = i_v d\eta + \eta(v) \eta$$

For a Hamiltonian function H on M we define the Hamiltonian vector field by

$$\bar{b}(X_H) = dH - (\mathcal{R}(H) + H)\eta$$

In Darboux coordinates we get this local expression

$$X_H = \frac{\partial H}{\partial p_i} \frac{\partial}{\partial q^i} - \left(\frac{\partial H}{\partial q^i} + p_i \frac{\partial H}{\partial z} \right) \frac{\partial}{\partial p_i} + \left(p_i \frac{\partial H}{\partial p_i} - H \right) \frac{\partial}{\partial z} \quad (8)$$

Therefore, an integral curve $(q^i(t), p_i(t), z(t))$ of X_H satisfies the contact Hamilton equations

$$\frac{dq^i}{dt} = \frac{\partial H}{\partial p_i} \quad (9)$$

$$\frac{dp_i}{dt} = -\left(\frac{\partial H}{\partial q^i} + p_i \frac{\partial H}{\partial z} \right) \quad (10)$$

$$\frac{dz}{dt} = \left(p_i \frac{\partial H}{\partial p_i} - H \right) \quad (11)$$

Example

Consider a Hamiltonian system given by the Hamiltonian

$$H(q, p, z) = \frac{p^2}{2m} + V(q) + \gamma z$$

where γ is a constant.

We obtain the following dynamical equations

$$\begin{aligned}\dot{q} &= \frac{p}{m} \\ \dot{p} &= -\frac{\partial V}{\partial q} - \gamma z \\ \dot{z} &= \frac{p^2}{2m} - V(q) - \gamma z\end{aligned}$$

that are just the damped Newtonian equations.

Symplectic Lagrangian formalism

Let $L : TQ \rightarrow \mathbb{R}$ be a Lagrangian function, where Q is a configuration n -dimensional manifold. Then, $L = L(q^i, \dot{q}^i)$, where (q^i) are coordinates in Q and $(q^i, \dot{q}^i)$ are the induced bundle coordinates in TQ . We will assume that L is regular, that is, the Hessian matrix

$$\left(\frac{\partial^2 L}{\partial \dot{q}^i \partial \dot{q}^j} \right)$$

is regular. Using the canonical endomorphism S on TQ locally defined by

$$S = dq^i \otimes \frac{\partial}{\partial \dot{q}^i}$$

one can construct a 1-form λ_L defined by

$$\lambda_L = S^*(dL)$$

and the 2-form

$$\omega_L = -d\lambda_L$$

Then, ω_L is symplectic if and only if L is regular.

Consider now the vector bundle isomorphism

$$\flat_L : T(TQ) \longrightarrow T^*(TQ)$$
$$\flat_L(v) = i_v \omega_L$$

and the Hamiltonian vector field

$$\xi_L = X_{E_L}$$

defined by

$$\flat_L(\xi_L) = dE_L$$

where $E_L = \Delta(L) - L$ is the energy. The vector field ξ_L (the Euler-Lagrange vector field) is locally given by

$$\xi_L = \dot{q}^i \frac{\partial}{\partial q^i} + B^i \frac{\partial}{\partial \dot{q}^i} \quad (12)$$

where

$$B^i \frac{\partial}{\partial \dot{q}^i} \left(\frac{\partial L}{\partial \dot{q}^j} \right) + \dot{q}^i \frac{\partial}{\partial q^i} \left(\frac{\partial L}{\partial \dot{q}^j} \right) - \frac{\partial L}{\partial q^j} = 0 \quad (13)$$

Now, if $(q^i(t), \dot{q}^i(t))$ is an integral curve of ξ_L , then it satisfies the usual Euler-Lagrange equations

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}^i} \right) - \frac{\partial L}{\partial q^i} = 0 \quad (14)$$

Cosymplectic Lagrangian formalism

We will recall here the geometric formalism for time-dependent Lagrangian systems, just to show the differences with the previous contact formalism. In this case, we also have a regular Lagrangian $L : TQ \times \mathbb{R} \longrightarrow \mathbb{R}$, but instead to consider the contact 1-form η_L we will consider the cosymplectic structure given by the pair (Ω_L, dz) , where

$$\Omega_L = -d\lambda_L$$

It is easy to check that, indeed, if L is regular then

$$dz \wedge \Omega_L^n \neq 0,$$

and conversely. Again, we have a Reeb vector field

$$\mathcal{R} = \frac{\partial}{\partial z} - W^{ij} \frac{\partial^2 L}{\partial \dot{q}^j \partial z} \frac{\partial}{\partial \dot{q}^i}$$

Consider now the following vector fields determined by means of the vector bundle isomorphism

$$\begin{aligned}\widetilde{b}_L &: T(TQ \times \mathbb{R}) \longrightarrow T^*(TQ \times \mathbb{R}) \\ \widetilde{b}_L(v) &= i_v \Omega_L + dz(v) dz\end{aligned}$$

say,

- ① the gradient vector field

$$\text{grad}(E_L) = \widetilde{\sharp}_L(dE_L)$$

- ② the Hamiltonian vector field

$$X_{E_L} = \mathcal{E}_L - \mathcal{R}(E_L) \mathcal{R}$$

- ③ and the evolution vector field

$$\mathcal{E}_L = X_{E_L} + \mathcal{R}$$

where $\widetilde{\sharp}_L = (\widetilde{b}_L)^{-1}$ is the inverse of $\widetilde{b}_L$.

The evolution vector field $\mathcal{E}_L$ is locally given by

$$\mathcal{E}_L = \dot{q}^i \frac{\partial}{\partial q^i} + B^i \frac{\partial}{\partial \dot{q}^i} + \frac{\partial}{\partial z} \quad (15)$$

where

$$B^i \frac{\partial}{\partial \dot{q}^i} \left(\frac{\partial L}{\partial \dot{q}^j} \right) + \dot{q}^i \frac{\partial}{\partial q^i} \left(\frac{\partial L}{\partial \dot{q}^j} \right) - \frac{\partial L}{\partial q^j} = 0 \quad (16)$$

Now, if $(q^i(t), \dot{q}^i(t), z(t))$ is an integral curve of $\mathcal{E}_L$ then it satisfies the usual Euler-Lagrange equations

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}^i} \right) - \frac{\partial L}{\partial q^i} = 0 \quad (17)$$

since $z = t + \text{constant}$.

F. Cantrijn, M. de León, E.A. Lacomba: Gradient vector fields on cosymplectic manifolds. *Journal of Physics A: Mathematical and General* 25 (1), 175–188.

Contact Lagrangian formalism

Let $L : TQ \times \mathbb{R} \longrightarrow \mathbb{R}$ be a Lagrangian function, where Q is a configuration n -dimensional manifold. Then, $L = L(q^i, \dot{q}^i, z)$, where (q^i) are coordinates in Q , $(q^i, \dot{q}^i)$ are the induced bundle coordinates in TQ and z is a global coordinate in $\mathbb{R}$.

We will assume that L is regular, that is, the Hessian matrix

$$\left(\frac{\partial^2 L}{\partial \dot{q}^i \partial \dot{q}^j} \right)$$

is regular.

From L , and using the canonical endomorphism S on TQ locally defined by

$$S = dq^i \otimes \frac{\partial}{\partial \dot{q}^i}$$

one can construct a 1-form λ_L defined by

$$\lambda_L = S^*(dL)$$

where now S and S^* are the natural extension of S and its adjoint operator S^* to $TQ \times \mathbb{R}$.

Therefore, we have

$$\lambda_L = \frac{\partial L}{\partial \dot{q}^i} dq^i$$

Now, the 1-form

$$\eta_L = dz - \frac{\partial L}{\partial \dot{q}^i} dq^i$$

is a contact form on $TQ \times \mathbb{R}$ if and if L is regular; indeed, if L is regular, then

$$\eta_L \wedge (d\eta_L)^n \neq 0,$$

and conversely. From now on, we always assume that it is the case. The corresponding Reeb vector field is

$$\mathcal{R} = \frac{\partial}{\partial z} - W^{ij} \frac{\partial^2 L}{\partial \dot{q}^j \partial z} \frac{\partial}{\partial \dot{q}^i},$$

where (W^{ij}) is the inverse matrix of the Hessian (W_{ij}) . The energy of the systems is defined by

$$E_L = \Delta(L) - L$$

where $\Delta = \dot{q}^i \frac{\partial}{\partial \dot{q}^i}$ is the Liouville vector field on TQ extended in the usual way to $TQ \times \mathbb{R}$. Therefore,

$$E_L = \dot{q}^i \frac{\partial L}{\partial \dot{q}^i} - L$$

Denote by

$$\bar{b}_L : T(TQ \times \mathbb{R}) \longrightarrow T^*(TQ \times \mathbb{R})$$

the vector bundle isomorphism

$$\bar{b}_L(v) = i_v(d\eta_L) + (i_v\eta_L)\eta_L$$

given by the contact form η_L on $TQ \times \mathbb{R}$. We shall denote its inverse by

$$\bar{\sharp}_L = (\bar{b}_L)^{-1}.$$

Denote by $\bar{\xi}_L$ the unique vector field defined by the equation

$$\bar{b}_L(\bar{\xi}_L) = dE_L - (\mathcal{R}(E_L) + E_L)\eta_L \quad (18)$$

A direct computation from eq. (18) shows that $\bar{\xi}_L$ is locally given by

$$\bar{\xi}_L = \dot{q}^i \frac{\partial}{\partial q^i} + \mathcal{B}^i \frac{\partial}{\partial \dot{q}^i} + (L - \dot{q}^i \frac{\partial}{\partial z} (\frac{\partial L}{\partial \dot{q}^i})) \frac{\partial}{\partial z} \quad (19)$$

where the components $\mathcal{B}^i$ satisfy the equation

$$\mathcal{B}^i \frac{\partial}{\partial \dot{q}^i} (\frac{\partial L}{\partial \dot{q}^j}) + \dot{q}^i \frac{\partial}{\partial q^i} (\frac{\partial L}{\partial \dot{q}^j}) + (L - \dot{q}^i \frac{\partial}{\partial z} (\frac{\partial L}{\partial \dot{q}^i})) - \frac{\partial L}{\partial q^i} = \frac{\partial L}{\partial \dot{q}^i} \frac{\partial L}{\partial z} \quad (20)$$

Then, if $(q^i(t), \dot{q}^i(t), z(t))$ is an integral curve of $\bar{\xi}_L$, and substituting its values in eq. (20) we obtain

$$\ddot{q}^i \frac{\partial}{\partial \dot{q}^i} \left(\frac{\partial L}{\partial \dot{q}^j} \right) + \dot{q}^i \frac{\partial}{\partial q^i} \left(\frac{\partial L}{\partial \dot{q}^j} \right) + \dot{z} \frac{\partial}{\partial z} \left(\frac{\partial L}{\partial \dot{q}^j} \right) - \frac{\partial L}{\partial q^i} = \frac{\partial L}{\partial \dot{q}^i} \frac{\partial L}{\partial z}$$

which corresponds to the generalized Euler-Lagrange equations considered by G. Herglotz in 1930.

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}^i} \right) - \frac{\partial L}{\partial q^i} = \frac{\partial L}{\partial \dot{q}^i} \frac{\partial L}{\partial z} \quad (21)$$

G. Herglotz: Berührungstransformationen, Lectures at the University of Gottingen, Gottingen, 1930.

The Hamiltonian formalism

Let $H : T^*Q \times \mathbb{R} \longrightarrow \mathbb{R}$ be a Hamiltonian function, say $H = H(q^i, p_i, z)$ where (q^i, p_i, z) are bundle coordinates in $H : T^*Q \times \mathbb{R}$. Consider the 1-form

$$\eta_Q = dz - \theta_Q$$

where θ_Q is the canonical Liouville form on T^*Q and we are considering the usual identifications for a form on T^*Q or $\mathbb{R}$ and its pull-back to $T^*Q \times \mathbb{R}$. In local coordinates, we have

$$\eta_Q = dz - p_i dq^i$$

So, η_Q is a contact form on $T^*Q \times \mathbb{R}$ and (q^i, p_i, z) are Darboux coordinates- Therefore, we can obtain a Hamiltonian vector field X_H which locally takes the same form that above.

Legendre transformation

Given a Lagrangian function $L : TQ \times \mathbb{R} \longrightarrow \mathbb{R}$ we can define the Legendre transformation

$$FL : TQ \times \mathbb{R} \longrightarrow T^*Q \times \mathbb{R}$$

given by

$$FL(q^i, \dot{q}^i, z) = (q^i, \hat{p}_i, z)$$

where

$$\hat{p}_i = \frac{\partial L}{\partial \dot{q}^i}$$

A direct computation shows that

$$FL^* \eta_Q = \eta_L$$

and then we have

$$T(FL)(\tilde{\xi}_L) = X_H$$

and consequently the generalized Euler-Lagrange equations are transformed into the contact Hamilton equations.

Variational formulation of contact Lagrangian mechanics: Herglotz principle

Variational formulation of contact Lagrangian mechanics

Let $L : TQ \times \mathbb{R} \rightarrow \mathbb{R}$ be a Lagrangian function. We will recall the so-called Herglotz's principle, a modification of Hamilton's principle that allows us to obtain Herglotz's equations, sometimes called generalized Euler-Lagrange equations.

Fix $q_1, q_2 \in Q$ and an interval $[a, b] \subset \mathbb{R}$. We denote by $\Omega(q_1, q_2, [a, b]) \subseteq (C^\infty([a, b] \rightarrow Q))$ the space of smooth curves ξ such that $\xi(a) = q_1$ and $\xi(b) = q_2$. This space has the structure of an infinite dimensional smooth manifold whose tangent space at ξ is given by the set of vector fields over ξ that vanish at the endpoints, that is,

$$\begin{aligned} T_\xi \Omega(q_1, q_2, [a, b]) = \{v_\xi \in C^\infty([a, b] \rightarrow TQ) \mid \\ \tau_Q \circ v_\xi = \xi, v_\xi(a) = 0, v_\xi(b) = 0\}. \end{aligned} \quad (22)$$

We will consider the following maps. Fix $c \in \mathbb{R}$. Let

$$\mathcal{Z} : \Omega(q_1, q_2, [a, b]) \rightarrow \mathcal{C}^\infty([a, b] \rightarrow \mathbb{R}) \quad (23)$$

be the operator that assigns to each curve ξ the curve $\mathcal{Z}(\xi)$ that solves the following ODE:

$$\frac{d\mathcal{Z}(\xi)(t)}{dt} = L(\xi(t), \dot{\xi}(t), \mathcal{Z}(\xi)(t)), \quad \mathcal{Z}(\xi)(a) = c. \quad (24)$$

Now we define the *action functional* as the map which assigns to each curve the solution to the previous ODE evaluated at the endpoint:

$$\begin{aligned} \mathcal{A} : \Omega(q_1, q_2, [a, b]) &\rightarrow \mathbb{R}, \\ \xi &\mapsto \mathcal{Z}(\xi)(b), \end{aligned} \quad (25)$$

that is, $\mathcal{A} = ev_b \mathcal{Z}$, where $ev_b : \zeta \mapsto \zeta(b)$ is the evaluation map at b .

Theorem(Contact variational principle) Let $L : TQ \times \mathbb{R} \rightarrow \mathbb{R}$ be a Lagrangian function and let $\xi \in \Omega(q_1, q_2, [a, b])$ be a curve in Q . Then, $(\xi, \dot{\xi}, \mathcal{Z}(\xi))$ satisfies the Herglotz's equations if and only if ξ is a critical point of $\mathcal{A}$. This theorem generalizes Hamilton's Variational Principle.

In the case that the Lagrangian is independent of the $\mathbb{R}$ coordinate (i.e., $L(x, y, z) = \hat{L}(x, y)$) the contact Lagrange equations reduce to the usual Euler-Lagrange equations. In this situation, we can integrate the ODE of (25) and we get

$$\mathcal{A}(\xi) = \int_a^b \hat{L}(\xi(t), \dot{\xi}(t)) dt + \frac{c}{b-a}, \quad (26)$$

that is, the usual Euler-Lagrange action up to a constant.

Contact manifolds and Jacobi manifolds

A Jacobi manifold is a triple (M, Λ, E) , where Λ is a bivector field (a skew-symmetric contravariant 2-tensor field) and $E \in \mathfrak{X}(M)$ is a vector field, so that the following identities are satisfied:

$$[\Lambda, \Lambda] = 2E \wedge \Lambda, \quad \mathcal{L}_E \Lambda = [E, \Lambda] = 0,$$

where $[\cdot, \cdot]$ is the Schouten-Nijenhuis bracket.

Given a Jacobi manifold (M, Λ, E) , we define the *Jacobi bracket*:

$$\begin{aligned} \{\cdot, \cdot\} : C^\infty(M) \times C^\infty(M) &\mapsto \mathbb{R}, \\ (f, g) &\mapsto \{f, g\}, \end{aligned}$$

where

$$\{f, g\} = \Lambda(df, dg) + fE(g) - gE(f).$$

This bracket is bilinear, antisymmetric, and satisfies the Jacobi identity. Furthermore it fulfills the weak Leibniz rule:

$$\text{supp}(\{f, g\}) \subseteq \text{supp}(f) \cap \text{supp}(g).$$

That is, $(C^\infty(M), \{\cdot, \cdot\})$ is a local Lie algebra in the sense of Kirillov. Conversely, given a local Lie algebra $(C^\infty(M), \{\cdot, \cdot\})$, we can find a Jacobi structure on M such that the Jacobi bracket coincides with the algebra bracket.

The weak Leibniz rule is equivalent to this identity:

$$\{f, gh\} = g\{f, h\} + h\{f, g\} + ghE(h)$$

Given a contact manifold (M, η) we can define a Jacobi structure (M, Λ, E) by

$$\Lambda(\alpha, \beta) = -d\eta(\sharp\alpha, \sharp\beta), \quad E = -\mathcal{R},$$

where $\sharp = \flat^{-1}$.

Poisson manifolds

One important particular case of Jacobi manifolds are Poisson manifolds (when $E = 0$). The corresponding Poisson bracket satisfies the following Leibniz rule

$$\{f, gh\} = \{f, g\}h + g\{f, h\}.$$

Examples of Poisson manifolds are symplectic and cosymplectic manifolds.

Cosymplectic manifold

A cosymplectic manifold is given by a triple (M, Ω, η) where M is a $(2n + 1)$ -dimensional manifold, Ω is an exact 2-form and η is an exact 1-form.

We consider the isomorphism

$$\begin{aligned}\tilde{b} : TM &\rightarrow T_x^*M \\ X &\mapsto i_X\Omega + \eta(X)\eta.\end{aligned}$$

If we denote its inverse by $\tilde{sharp} = \tilde{b}^{-1}$, then

$$\Lambda(\alpha, \beta) = \Omega(\tilde{\sharp}\alpha, \tilde{\sharp}\beta),$$

is a Poisson tensor on M .

Locally conformal symplectic manifolds

An almost symplectic manifold is said to be *locally conformally symplectic* if for each point $x \in M$ there is an open neighborhood U such that $d(e^\sigma \Omega) = 0$, for $\sigma : U \rightarrow \mathbb{R}$, so $(U, e^\sigma \Omega)$ is a symplectic manifold. If $U = M$, then it is said to be globally conformally symplectic.

One can see that these local 1-forms $d\sigma$ defines a closed 1-form θ such that

$$d\Omega = \theta \wedge \Omega.$$

The one-form θ is called the *Lee one-form*. Locally conformally symplectic manifolds (L.C.S.) with Lee form $\theta = 0$ are symplectic manifolds. We define a bivector Λ on M and a vector field E given by

$$\Lambda(\alpha, \beta) = \Omega(b^{-1}(\alpha), b^{-1}(\beta)) = \Omega(\sharp(\alpha), \sharp(\beta)), \quad E = b^{-1}(\theta)$$

with $\alpha, \beta \in \Omega^1(M)$ and $b : \mathfrak{X}(M) \rightarrow \Omega^1(M)$ is the isomorphism of $C^\infty(M)$ modules defined by $b(X) = \iota_X \Omega$. Here $\sharp = b^{-1}$. The vector field E satisfies $\iota_E \theta = 0$ and $\mathcal{L}_E \Omega = 0, \mathcal{L}_E \theta = 0$. Then, (M, Λ, E) is an even dimensional Jacobi manifold.

Let (M, Λ, E) be a Jacobi manifold. We define the following morphism of vector bundles:

$$\begin{aligned} \sharp_{\Lambda} : TM^* &\rightarrow TM \\ \alpha &\mapsto \Lambda(\alpha, \cdot), \end{aligned}$$

which also induces a morphism of $C^{\infty}(M)$ -modules between 1-forms and vector fields.

In the case of a contact manifold, this is given by

$$\sharp_{\Lambda} \alpha = \sharp \alpha - \alpha(\mathcal{R})\mathcal{R},$$

since

$$\eta(\sharp_{\Lambda} \alpha) = \alpha(\mathcal{R})$$

for any 1-form α .

For a contact manifold, $\sharp_{\Lambda}$ is not an isomorphism. In fact, $\ker \sharp_{\Lambda} = \langle \eta \rangle$ and $\operatorname{Im} \sharp_{\Lambda} = \mathcal{H}$.

Vector fields associated with functions f on the algebra of smooth functions $C^\infty(M)$ are defined as

$$X_f = \sharp_\Lambda(df) + fE,$$

The *characteristic distribution* $\mathcal{C}$ of (M, Λ, E) is generated by the values of all the vector fields X_f . This characteristic distribution $\mathcal{C}$ is defined in terms of Λ and E as follows

$$\mathcal{C}_p = \sharp_{\Lambda_p}(T_p^*M) + \langle E_p \rangle, \quad \forall p \in M$$

where $\sharp_p : T_p^*M \rightarrow T_pM$ is the restriction of $\sharp_\Lambda$ to T_p^*M for every $p \in M$. Then, $\mathcal{C}_p = \mathcal{C} \cap T_pM$ is the vector subspace of T_pM generated by E_p and the image of the linear mapping $\sharp_p$.

The distribution is said to be *transitive* if the characteristic distribution is the whole tangent bundle TM .

Structure theorem for Jacobi manifolds

Theorem

The characteristic distribution of a Jacobi manifold (M, Λ, E) is completely integrable in the sense of Stefan–Sussmann, thus M defines a foliation whose leaves are not necessarily of the same dimension, and it is called the characteristic foliation. Each leaf has a unique transitive Jacobi structure such that its canonical injection into M is a Jacobi map (that is, it preserves the Jacobi brackets). Each can be

- ① *A locally conformally symplectic (or a symplectic) manifold if the dimension is even.*
- ② *A manifold equipped with a contact one-form if its dimension is odd.*

Submanifolds of a contact manifold

As in the case of symplectic manifolds, we can consider several interesting types of submanifolds of a contact manifold (M, η) . To define them, we will use the following notion of *complement* for contact structures: Let (M, η) be a contact manifold and $x \in M$. Let $\Delta_x \subset T_x M$ be a linear subspace. We define the *contact complement* of Δ_x

$$\Delta_x^{\perp_\Lambda} = \sharp_\Lambda(\Delta_x^\circ),$$

where $\Delta_x^\circ = \{\alpha_x \in T_x^* M \mid \alpha_x(\Delta_x) = 0\}$ is the annihilator.

We extend this definition for distributions $\Delta \subseteq TM$ by taking the complement pointwise in each tangent space.

Let $N \subseteq M$ be a submanifold. We say that N is:

- *Isotropic* if $TN \subseteq TN^{\perp\lambda}$.
- *Coisotropic* if $TN \supseteq TN^{\perp\lambda}$.
- *Legendrian* if $TN = TN^{\perp\lambda}$.

Let (M, η) be a contact manifold of dimension $2n + 1$.

One can easily prove the following characterization of a Legendrian submanifold:

A submanifold N of M is Legendrian if and only if it is a maximal integral manifold of $\ker \eta$ (and then it has dimension n).

Proposition A section $\gamma : Q \longrightarrow T^*Q \times \mathbb{R}$ of the canonical projection $T^*Q \times \mathbb{R} \longrightarrow Q$ is a Legendre submanifold of $(T^*Q \times \mathbb{R}, \eta_Q)$ if and only if γ is locally the 1-jet of a function $f : Q \longrightarrow \mathbb{R}$.

The above result is the natural extension of the well-known fact that a section σ of the cotangent bundle $\pi_Q : T^*Q \longrightarrow Q$ is a Lagrangian submanifold with respect to the canonical symplectic structure $\omega_Q = -d\theta_Q$ on T^*Q if and only if σ is a closed 1-form (and hence, locally exact).

Other topics

In the next slides I will indicate some of the research topics we have been working on in recent years. I will indicate only the topic, a very brief description and the references where you can find more details.

Contact Hamiltonian dynamics and Legendrian submanifolds

Theorem (Contactification of the tangent bundle)

Let (M, η) be a contact manifold. Let $\bar{\eta}$ be a one form on $TM \times \mathbb{R}$ such that

$$\bar{\eta} = \eta^C + t\eta^V,$$

where t is the usual coordinate on $\mathbb{R}$ and η^C and η^V are the complete and vertical lifts of η to TM . Then, $(TM \times \mathbb{R}, \bar{\eta})$ is a contact manifold with Reeb vector field $\bar{\mathcal{R}} = \mathcal{R}^V$.

Theorem

Let (M, η, H) be a contact Hamiltonian system. Then

$$(X_H \times (\mathcal{R}(H))) \subseteq (TM \times \mathbb{R}, \bar{\eta})$$

is a Legendrian submanifold.

The result follows since

$$\mathcal{L}_{X_H} \eta = -\mathcal{R}(H)\eta$$

Coisotropic reduction in contact manifolds

Theorem

Let $\iota : N \longrightarrow M$ be a coisotropic submanifold. Then $TN^{\perp\wedge}$ is involutive. If the quotient $\tilde{N} = TN/TN^{\perp\wedge}$ is a manifold and N does not have horizontal points, let $\pi : N \rightarrow \tilde{N}$ be the projection. Then there exists a unique 1-form $\tilde{\eta}$ on $\tilde{N}$ such that $\tilde{\eta} = \pi_*(\eta)$ and $(N, \tilde{\eta})$ is a contact manifold.

Furthermore, if N consists only of vertical points, then $\tilde{\mathcal{R}} = \pi^*\mathcal{R}$ is well defined and is the correspondig Reeb vector field.

Corollary

With the notations from previous theorem, assume that $L \subseteq M$ is Legendrian, N does not have horizontal points, and N and L have clean intersection (that is, $N \cap L$ is a submanifold and $T(N \cap L) = TN \cap TL$). Then $\tilde{L} = \pi(L) \subseteq \tilde{N}$ is Legendrian.

M. de León, M. Lainz-Valcázar: Contact Hamiltonian systems. *J. Math. Phys.* **60** (2019), no. 10, 102902, 18 pp.

Infinitesimal symmetries and dissipate quantities

M. de León, M. Lainz-Valcázar: Infinitesimal symmetries in contact Hamiltonian systems. *J. Geom. Phys.* **153** (2020), 103651, 13 pp

- 1 In this paper, we extend the well-known Noether theorem for Lagrangian systems to contact Lagrangian systems.
- 2 We introduce a classification of infinitesimal symmetries and obtain the corresponding dissipated quantities.
- 3 We notice that in contact dynamics, the existence of infinitesimal symmetries does not produce conserved quantities, but functions that dissipate at the same rate than the energy; so, the corresponding quotients are true conserved quantities.
- 4 The case of contact reduction by a Lie group of symmetries is also discussed.

Contact nonholonomic mechanical systems

M. de León, V.M. Jiménez, M. Lainz-Valcázar: Contact Hamiltonian and Lagrangian systems with nonholonomic constraints. *Journal of Geometric Mechanics* (2021) doi: 10.3934/jgm.2021001.

- In this article we develop a theory of contact systems with nonholonomic constraints.
- We obtain the dynamics from Herglotz's variational principle, by restricting the variations so that they satisfy the nonholonomic constraints.
- We prove that the nonholonomic dynamics can be obtained as a projection of the unconstrained Hamiltonian vector field.
- Finally, we construct the nonholonomic bracket, which is an almost Jacobi bracket on the space of observables and provides the nonholonomic dynamics.

Singular contact Lagrangian systems

M. de León, M. Lainz-Valcázar: Singular Lagrangians and precontact Hamiltonian systems. *Int. J. Geom. Methods Mod. Phys.* **16** (2019), no. 10, 1950158, 39 pp.

- In this paper we discuss singular Lagrangian systems on the framework of contact geometry. These systems exhibit a dissipative behavior in contrast with the symplectic scenario.
- We develop a constraint algorithm similar to the presymplectic one studied by Gotay and Nester (the geometrization of the well-known Dirac-Bergman algorithm).
- We also construct the Hamiltonian counterpart and prove the equivalence with the Lagrangian side.
- A Dirac-Jacobi bracket is constructed similar to the Dirac bracket.

Contact discrete dynamics

A.A. Simoes, D. Martín de Diego, M. de León, M. Lainz-Valcázar: On the geometry of discrete contact mechanics. *arXiv preprint arXiv:2003.11892* (2020).

- we introduce a discrete Herglotz Principle and the corresponding discrete Herglotz Equations for a discrete Lagrangian in the contact setting. This allows us to develop convenient numerical integrators for contact Lagrangian systems that are conformally contact by construction.
- The existence of an exact Lagrangian function is also discussed.

Uniform formalism

M. de León, J. Gaset, M. Lainz-Valcázar, X. Rivas, N. Román-Roy:
Unified Lagrangian-Hamiltonian formalism for contact systems. *Fortschr. Phys.* **68** (2020), no. 8, 2000045, 12 pp.

- We develop a unified geometric framework for describing both the Lagrangian and Hamiltonian formalisms of contact autonomous mechanical systems, which is based on the approach of the pionnering work of R. Skinner and R. Rusk.
- This framework permits to skip the second order differential equation problem, which is obtained as a part of the constraint algorithm (for singular or regular Lagrangians), and is specially useful to describe singular Lagrangian systems.
- Some examples are also discussed to illustrate the method.

Contact Optimal Control Theory

M. de León, M. Lainz-Valcázar, M.C. Muñoz-Lecanda: Optimal control, contact dynamics and Herglotz variational problem *arXiv preprint arXiv:2006.14326* (2020)

- We combine two main topics in mechanics and optimal control theory: contact Hamiltonian systems and Pontryagin Maximum Principle.
- As an important result, a contact Pontryagin Maximum Principle that permits to deal with optimal control problems with dissipation is developed.
- Also, the Herglotz optimal control problem is stated, in such a way that generalizes simultaneously the Herglotz variational principle and an optimal control problem.
- Some applications to the study of a thermodynamic system are provided.

Existence of invariant measures

A. Bravetti, M. de León, J.C. Marrero, E. Padrón: Invariant measures for contact Hamiltonian systems: symplectic sandwiches with contact bread
Journal of Physics A: Mathematical and Theoretical **53** (45), (2020) 455205.

- An important topic in dynamical systems is the existence of invariant measures. We prove that, under some natural conditions, Hamiltonian systems on a contact manifold C can be split into a Reeb dynamics on an open subset of C and a Liouville dynamics on a submanifold of C of codimension 1.
- Thus, an invariant measure is obtained for the Reeb dynamics, and moreover, under certain completeness conditions, the existence of an invariant measure for the Liouville dynamics can be characterized using the notion of a symplectic sandwich with contact bread developed in this paper.

Contact higher order mechanics

M. de León, J. Gaset, M Lainz-Valcázar, M.C. Muñoz-Lecanda, N. Román-Roy: Higher-order contact mechanics. *Annals of Physics* **425** 168396 (2021)

- We present a complete theory of higher-order autonomous contact mechanics, which allows us to describe higher-order dynamical systems with dissipation.
- The essential tools for the theory are the extended higher-order tangent bundles, $T^k Q \times \mathbb{R}$, and its canonical geometric structures. This allow us to state the Lagrangian and Hamiltonian formalisms for these kinds of systems, as well as their variational formulation.
- In that paper, a unified description that encompasses the Lagrangian and Hamiltonian equations as well as their relationship through the Legendre map; all of them are obtained from the contact dynamical equations and the constraint algorithm that is implemented because, in this formalism, the dynamical systems are always singular.
- The theory is applied to some interesting examples.

Hamilton-Jacobi theory

M. de León, C. Sardón: Cosymplectic and contact structures for time-dependent and dissipative Hamiltonian systems. *J. Phys. A* **50** (2017), no. 25, 255205, 23pp.

M. de León, M. Lainz-Valcázar, A. Muñoz-Brea: The Hamilton–Jacobi theory for contact Hamiltonian systems. Preprint (2021).

- 1 In the first paper, we discuss a comparison between the different Hamilton-Jacobi equations in these three different contexts: symplectic, cosymplectic and contact systems.
- 2 In the second paper, we concentrate our study on the case of contact Hamiltonian mechanics, and obtain four different Hamilton-Jacobi equations, considering the Hamiltonian vector field and the evolution vector field, separately. In addition, we establish the relation with the symplectified system.

PART II: Applications to thermodynamics

Consider a simple thermodynamic system, for instance, gas in a confined compartment with volume V and pressure P at temperature T .

The state properties of the gas are described by a 2-dimensional submanifold of the thermodynamic phase space, $\mathbb{R}^5$, with coordinates:

- E energy
- S entropy
- V, P , and T

This submanifold is a Legendre submanifold of the contact manifold $(\mathbb{R}^5, \eta)$, where

$$\eta = dE - TdS + PdV$$

Here, $E = E(S, V)$ and the equations of the Legendre submanifold are

$$T = \frac{\partial E}{\partial S}, \quad -P = \frac{\partial E}{\partial V}$$

E, S, V are called **extensive** variables

T, P are called **intensive** variables

Applications to thermodynamics

A.A. Simoes, D. Martín de Diego, M. de León, M. Lainz-Valcázar:
Contact geometry for simple thermodynamical systems with friction.
*Proceedings of the Royal Society A: Mathematical, Physical and
Engineering Sciences* **476** 2241 (2020) doi.org/10.1098/rspa.2020.0244.

A.A. Simoes, D. Martín de Diego, M. de León, M. Lainz-Valcázar: The
geometry of some thermodynamic systems. arXiv preprint
arXiv:2012.07404 (to appear in an Springer book, 2021).

In this second part, we will introduce a differential geometric framework that incorporates in a very natural way fundamental thermodynamical concepts as the free energy and the rate of entropy production.

Typically, in the previous literature, this description needs to introduce appropriate Poisson and dissipation brackets with combined properties that allows the two laws of thermodynamics to be satisfied.

One of the most successful methods are based on the introduction of *metriplectic structures*:

Allan N. Kaufman. Dissipative Hamiltonian systems: a unifying principle. Phys. Lett. A, 100(8):419–422, 1984.

Philip J. Morrison. A paradigm for joined Hamiltonian and dissipative systems. volume 18, pages 410–419. 1986. Solitons and coherent structures (Santa Barbara, Calif., 1985).

coupling a Poisson and a gradient structure, where the entropy S is now constructed from a Casimir function of the Poisson structure.

Other approaches like in

B. J. Edwards and A. N. Beris. Noncanonical Poisson bracket for nonlinear elasticity with extensions to viscoelasticity. J. Phys. A, 24(11):2461–2480, 1991.

B. J. Edwards and A. N. Beris. Noncanonical Poisson bracket for nonlinear elasticity with extensions to viscoelasticity. J. Phys. A, 24(11):2461–2480, 1991.

use similar techniques, called *single generation formalism* introducing a generalized bracket which is naturally divided into two parts: a non-canonical Poisson bracket and a new dissipation bracket.

The derived structures are capable of reproducing both reversible and irreversible evolutions providing a unifying formalism for many systems ruled by the laws of thermodynamics.

These approaches have proved to be very useful for the description of complex thermodynamical systems and also facilitate their numerical integration.

Recently, Gay-Balmaz and Yoshimura

F. Gay-Balmaz and H. Yoshimura. A Lagrangian variational formulation for nonequilibrium thermodynamics. Part I: Discrete systems. *Journal of Geometry and Physics*, 111:169–193, January 2017.

F. Gay-Balmaz and H. Yoshimura. From Lagrangian Mechanics to Nonequilibrium Thermodynamics: A Variational Perspective. *Entropy*, 21(1):8, January 2019.

have introduced a variational principle for the description of thermodynamical systems.

Their formulation extends the Hamilton principle of classical mechanics to include irreversible processes by introducing additional phenomenological and variational constraints.

A more geometrical approach is based on the use of contact geometry; indeed, the thermodynamical phase space is equipped with a contact structure:

- Using the contact structure, it is possible to associate to each function f , a Hamiltonian vector field X_f which is the infinitesimal generator of a contact transformation.
- In this framework, the manifold of equilibrium states is represented by a Legendre submanifold N and the Hamiltonian vector field X_f is tangent to N if and only if the function f vanishes on N , that is, the Legendre submanifold is contained on the zero level set of the Hamiltonian function.
- The flow of X_f restricted to the Legendrian submanifold is interpreted as thermodynamical processes.

R. Mrugala, J. D. Nulton, J. Ch. Schon, and P. Salamon. Contact structure in thermodynamic theory. Reports on Mathematical Physics, 29(1):109–121, February 1991.

R. Mrugala. Continuous contact transformations in thermodynamics. In Proceedings of the XXV Symposium on Mathematical Physics (Torun, 1992), vol. 33, pages 149–154, 1993.

Another approach to the dynamics of thermodynamical processes is the one used in

R. Balian and P. Valentin. Hamiltonian structure of thermodynamics with gauge. The European Physical Journal B-Condensed Matter and Complex Systems, 21(2):269–282, 2001.

A. Van der Schaft and B. Maschke. Geometry of thermodynamic processes. Entropy, 20(12):925, 2018.

which is based on homogeneous symplectic Hamiltonian systems, and is completely equivalent to the contact Hamiltonian vector field approach.

A recent paper is

Arjan van der Schaft: Liouville geometry of classical thermodynamics.
arXiv:2102.05493

Contact geometry and contact dynamics

We will consider some ingredients of contact geometry that we will need in the sequel.

Let M be a differentiable manifold of dimension $2n + 1$ and a 1-form η on M . We say that η is a contact 1-form if $\eta \wedge (d\eta)^n \neq 0$ at every point. Then, we call (M, η) a contact manifold. A distinguished vector field for a contact manifold is the Reeb vector field $\mathcal{R} \in \mathfrak{X}(M)$ univocally characterized by

$$i_{\mathcal{R}}\eta = 1 \quad \text{and} \quad i_{\mathcal{R}}d\eta = 0 .$$

We can define also an isomorphism of $C^\infty(M, \mathbb{R})$ modules by

$$\begin{aligned} \bar{b}: \mathfrak{X}(M) &\longrightarrow \Omega^1(M) \\ X &\longmapsto i_X d\eta + \eta(X)\eta \end{aligned}$$

Observe that $\bar{b}^{-1}(\eta) = \mathcal{R}$.

Using the generalized Darboux theorem, we have canonical coordinates (q^i, p_i, S) , $1 \leq i \leq n$ in a neighborhood of every point $x \in M$, such that the contact 1-form η and the Reeb vector field are:

$$\eta = dS - p_i dq^i \quad \text{and} \quad \mathcal{R} = \frac{\partial}{\partial S} .$$

Define the bi-vector Λ on M by

$$\Lambda(\alpha, \beta) = -d\eta(b^{-1}(\alpha), b^{-1}(\beta)), \quad \alpha, \beta \in \Omega^1(M) . \quad (27)$$

In canonical coordinates,

$$\Lambda = \frac{\partial}{\partial p_i} \wedge \left(\frac{\partial}{\partial q^i} + p_i \frac{\partial}{\partial S} \right) \quad (28)$$

Define the $C^\infty(M, \mathbb{R})$ -linear mapping

$$\sharp_\Lambda : \Omega^1(M) \rightarrow \mathfrak{X}(M)$$

by $\langle \beta, \sharp_\Lambda(\alpha) \rangle = \Lambda(\alpha, \beta)$ with $\alpha, \beta \in \Omega^1(M)$.

Given a function $f \in C^\infty(M, \mathbb{R})$ we will define the following vector fields

- **Hamiltonian or contact vector field** X_f defined by

$$X_f = \sharp_\Lambda(df) - f\mathcal{R}$$

or in other terms, X_f is the unique vector field such that

$$\bar{b}(X_f) = df - (\mathcal{R}(f) + f)\eta.$$

In canonical coordinates:

$$X_f = \frac{\partial f}{\partial p_i} \frac{\partial}{\partial q^i} - \left(\frac{\partial f}{\partial q^i} + p_i \frac{\partial f}{\partial S} \right) \frac{\partial}{\partial p_i} + \left(p_i \frac{\partial f}{\partial p_i} - f \right) \frac{\partial}{\partial S}$$

- **The evolution or horizontal vector field**

$$\mathcal{E}_f = \sharp_\Lambda(df) = X_f + f\mathcal{R}$$

or

$$\bar{b}(\mathcal{E}_f) = df - \mathcal{R}(f)\eta.$$

In canonical coordinates:

$$\mathcal{E}_f = \frac{\partial f}{\partial p_i} \frac{\partial}{\partial q^i} - \left(\frac{\partial f}{\partial q^i} + p_i \frac{\partial f}{\partial S} \right) \frac{\partial}{\partial p_i} + p_i \frac{\partial f}{\partial p_i} \frac{\partial}{\partial S}$$

Remarks

- 1 We will see that the evolution vector field will be useful to describe some simple isolated thermodynamical systems with friction, where the variable S will play the role of the entropy of the system.
- 2 The interpretation of the variable S as being the entropy of the system excludes the possibility of using cosymplectic geometry to describe thermodynamical systems. Indeed, if the thermodynamical equations were the integral curves of the cosymplectic Hamiltonian vector field, then the entropy production would be constant, which is not the general situation.

Jacobi and Cartan brackets

The pair $(\Lambda, E = -R)$ is a particular case of Jacobi structure since it satisfies

$$[\Lambda, \Lambda] = 2E \wedge \Lambda \quad \text{and} \quad [\Lambda, E] = 0.$$

From this Jacobi structure we can define the Jacobi bracket as follows:

$$\{f, g\} = \Lambda(df, dg) + fE(g) - gE(f), \quad f, g \in C^\infty(M, \mathbb{R})$$

The mapping $\{, \} : C^\infty(M, \mathbb{R}) \times C^\infty(M, \mathbb{R}) \longrightarrow C^\infty(M, \mathbb{R})$ is bilinear, skew-symmetric and satisfies the Jacobi's identity but, in general, it does not satisfy the Leibniz rule; this last property is replaced by a weaker condition:

$$\text{Supp } \{f, g\} \subset \text{Supp } f \cap \text{Supp } g.$$

In this sense, this bracket generalizes the well-known Poisson brackets. Indeed, a Poisson manifold is a particular case of Jacobi manifold.

In local coordinates

$$\{f, g\} = \frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q^i} - \frac{\partial f}{\partial q^i} \frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial S} \left(p_i \frac{\partial g}{\partial p_i} - g \right) + \frac{\partial g}{\partial S} \left(p_i \frac{\partial f}{\partial p_i} - f \right)$$

It is also interesting for us to introduce the *Cartan bracket* (that does not obey the Jacobi identity)

$$\begin{aligned}[f, g] &= \Lambda(df, dg) \\ &= \frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q^i} - \frac{\partial f}{\partial q^i} \frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial S} \left(p_i \frac{\partial g}{\partial p_i} \right) + \frac{\partial g}{\partial S} \left(p_i \frac{\partial f}{\partial p_i} \right)\end{aligned}$$

Mechanics

Our main example of contact manifold along this talk will be $T^*Q \times \mathbb{R}$, where Q is n -dimensional manifold, with contact structure defined by

$$\eta_Q = pr_2^*(dS) - pr_1^*(\theta_Q) \equiv dS - \theta_Q$$

where $pr_1 : T^*Q \times \mathbb{R} \rightarrow T^*Q$ and $pr_2 : T^*Q \times \mathbb{R} \rightarrow \mathbb{R}$ are the canonical projections and θ_Q is the Liouville 1-form on the cotangent bundle defined by

$$\eta_Q(X_{\mu_q}) = \langle \mu_q, T_{\mu_q} \pi_Q X_{\mu_q} \rangle$$

being $X_{\mu_q} \in T_{\mu_q} T^*Q$. Taking bundle coordinates (q^i, p_i) on T^*Q we have that $\eta_Q = dS - p_i dq^i$.

On such a manifold we can define the bi-vector

$$\Lambda_0 = \Lambda + \sharp_\Lambda(dS) \wedge \mathcal{R}$$

which is Poisson, that is $[\Lambda_0, \Lambda_0] = 0$. In coordinates,

$$\Lambda_0 = \frac{\partial}{\partial p_i} \wedge \frac{\partial}{\partial q^i}$$

is like the canonical Poisson bracket on T^*Q but now applied to functions on $T^*Q \times \mathbb{R}$.

Observe that in this case the Cartan bracket can be rewritten in terms of the Poisson bracket induced by Λ_0 and an extra term that describe the thermodynamical behaviour. That is,

$$[f, g] = \{f, g\}_{\Lambda_0} - \frac{\partial f}{\partial S} \Delta g + \frac{\partial g}{\partial S} \Delta f$$

where $\Delta = -\sharp_{\Lambda}(dS)$ is the Liouville vector field:

$$\Delta = p_i \frac{\partial}{\partial p_i}$$

We will denote by

$$\{f, g\}_{\Delta} = \frac{\partial g}{\partial S} \Delta f - \frac{\partial f}{\partial S} \Delta g$$

Then, the Cartan bracket is written as in the single generation formalism as

$$[f, g] = \{f, g\}_{\Lambda_0} + \{f, g\}_{\Delta} \quad (29)$$

Now, we will discuss some interesting properties of the qualitative behaviour of the evolution vector field $\mathcal{E}_f$.

Proposition The Lie derivative of the contact form η with respect to the evolution vector field $\mathcal{E}_f$ associated to the Hamiltonian function f satisfies the following relation

$$\mathcal{L}_{\mathcal{E}_f}\eta = -R(f)\eta + df .$$

Proof:

The proof is a trivial consequence of the properties of the Lie derivative and the properties of the Hamiltonian vector field:

$$\begin{aligned}\mathcal{L}_{\mathcal{E}_f}\eta &= \mathcal{L}_{X_f+fR}\eta = \mathcal{L}_{X_f}\eta + \mathcal{L}_{fR}\eta \\ &= -R(f)\eta + (i_R\eta)df = -R(f)\eta + df\end{aligned}$$

Theorem

Let $\mathcal{L}_c(f) = f^{-1}(c)$ be a level set of $f : M \rightarrow \mathbb{R}$ where $c \in \mathbb{R}$. We assume that $\mathcal{L}_c(f) \neq \emptyset$ and $R(f)(x) \neq 0$ for all $x \in \mathcal{L}_c(f)$. Then

- 1 The 2-form $\omega_c \in \Omega^2(\mathcal{L}_c(f))$ defined by

$$\omega_c = -di_c^* \eta$$

is an exact symplectic structure. Here $i_c : \mathcal{L}_c(f) \hookrightarrow M$ denotes the canonical inclusion

- 2 If Δ_c is the Liouville vector field, that is,

$$i_{\Delta_c} \omega_c = i_c^* \eta$$

then the restriction of $\mathcal{E}_f$ to $\mathcal{L}_c(f)$ verifies that

$$\mathcal{E}_f|_{\mathcal{L}_c(f)} = R(f)|_{\mathcal{L}_c(f)} \Delta_c$$

Proof:

The form ω_c is trivially closed. To see that it is a symplectic form, we just need to check that it is non degenerate. Let $p \in \mathcal{L}_c(f)$. Notice that, at that point, $\omega_c = -d\eta|_{T_p \mathcal{L}_c(f)}$. By the condition $R(f) \neq 0$, we have that R_p (and, hence $\ker \eta = \text{span } \langle R \rangle$) is transverse to $T_p \mathcal{L}_c(f)$. But since $\eta_p \wedge d\eta_p^n \neq 0$, then $d\eta|_V$ is non-degenerate for every subspace V transverse to $\ker \eta$. Therefore, ω_c is also non-degenerated.

For the second part, we first remark that $\mathcal{E}_f(f) = 0$, hence $(i_c)_* \mathcal{E}_f = \mathcal{E}_f|_{\mathcal{L}_c(f)}$ is a well-defined vector field. By the above Proposition and Cartan's identity

$$i_{\mathcal{E}_f} d\eta = -R(f)\eta + df.$$

Pulling back by i_c , we get

$$i_{(i_c)_* \mathcal{E}_f} i_c^* d\eta = -(R(f) \circ i_c) i_c^* \eta + di_c^* f = -(R(f) \circ i_c) i_c^* \eta,$$

dividing by $-(R(f) \circ i_c)$,

$$-i_{(i_c)_* \mathcal{E}_f / R(f)} i_c^* d\eta = i_{(i_c)_* \mathcal{E}_f / R(f)} \omega_c = i_c^* \eta.$$

Thus, $(i_c)_* (\mathcal{E}_f / R(f)) = \Delta_c$, as we wanted to show.

Remarks

- ① Observe that since

$$\mathcal{E}_f|_{\mathcal{L}_c(f)} = R(f)|_{\mathcal{L}_c(f)} \Delta_c$$

then, the dynamics on each energy level is like a Liouville dynamics after a time reparametrization

$$dt = \frac{1}{R(f)} d\tau .$$

- ② It is interesting to note that $T^*Q \times \mathbb{R}$ is also the phase space for time-dependent dynamics. In this case, the appropriate formalism is the cosymplectic formalism where the canonical cosymplectic structure is given by (dt, ω_Q)

A. Bravetti, M. de León, J. C. Marrero, E. Padrón: Invariant measures for contact Hamiltonian systems: symplectic sandwiches with contact bread
arXiv:2006.15123

- We prove that, under some natural conditions, Hamiltonian systems on a contact manifold C can be split into a Reeb dynamics on an open subset of C and a Liouville dynamics on a submanifold of C of codimension 1.
- For the Reeb dynamics we find an invariant measure.
- Moreover, we show that, under certain completeness conditions, the existence of an invariant measure for the Liouville dynamics can be characterized using the notion of a symplectic sandwich with contact bread.

Simple thermodynamical systems

We will use the evolution vector field to describe simple thermodynamical systems, that is, thermodynamical systems whose configuration space is composed by just one scalar thermal variable (in our case the entropy) and a finite set of mechanical variables (position and momenta). We will assume that the system is isolated, that is, there is not any transfer of work, matter or heat.

The isolated simple thermodynamical systems are described by a Lagrangian function:

$$\begin{aligned} L : \quad TQ \times \mathbb{R} &\longrightarrow \mathbb{R} \\ (v_q, S) &\longmapsto L(v_q, S) \end{aligned}$$

where Q is the configuration manifold describing the mechanical part of the thermodynamical system, TQ is the tangent bundle with canonical projection $\tau_Q : TQ \rightarrow Q$ given by $\tau_Q(v_q) = q$. The entropy of the system is described by the real variable $S \in \mathbb{R}$. If we consider coordinates (q^i) on Q and induced coordinates $(q^i, \dot{q}^i)$ on TQ , then $\tau_Q(q^i, \dot{q}^i) = (q^i)$.

We will see that the Lagrangian function itself will produce a friction force satisfying naturally the two laws of thermodynamics.

We will assume that the Lagrangian system is regular, that is, the matrix

$$(W_{ij}) = \left(\frac{\partial^2 L}{\partial \dot{q}^i \partial \dot{q}^j} \right)$$

is regular or, equivalently, the mapping $\mathbb{F}L : TQ \times \mathbb{R} \rightarrow T^*Q \times \mathbb{R}$ is a local diffeomorphism, where:

$$\mathbb{F}L(q^i, \dot{q}^i, S) = \left(q^i, \frac{\partial L}{\partial \dot{q}^i}, S \right)$$

is the Legendre transform. For simplicity, we will assume that the Legendre transform is a global diffeomorphism, since if it was only a local diffeomorphism we could proceed analogously by restricting to a neighbourhood. Then, we may define a Hamiltonian function $H : T^*Q \times \mathbb{R} \rightarrow \mathbb{R}$ given by

$$H(q^i, p_i, S) = p_i \dot{q}^i - L(q^i, \dot{q}^i, S)$$

where now the coordinates $\dot{q}^i$ are implicitly defined by the relations $p_j = \frac{\partial L}{\partial \dot{q}^j}(q^i, \dot{q}^i, S)$.

The equations of motion defined by the evolution vector field $\mathcal{E}_H$ are

$$\begin{aligned}\frac{dq^i}{dt} &= \frac{\partial H}{\partial p_i}, \\ \frac{dp_i}{dt} &= -\frac{\partial H}{\partial p_i} - p_i \frac{\partial H}{\partial S} \\ \frac{dS}{dt} &= p_i \frac{\partial H}{\partial p_i}.\end{aligned}$$

The vector field $\mathcal{E}_H$ satisfies the following two properties that correspond to the first and second laws of thermodynamics: conservation of the energy of an isolated system and irreversibility of the processes, that is, non-decreasing entropy production.

Proposition The integral curves of $\mathcal{E}_H$ satisfies the following properties:

- 1 $\mathcal{E}_H(H) = 0$, that is, $\frac{dH}{dt} = 0$;
- 2 $\mathcal{E}_H(S) = \Delta(H)$, that is, $\frac{dS}{dt} = \Delta H$.

Proof Both are a direct consequence of the definition of the evolution vector field $\mathcal{E}_H = \sharp_{\Lambda}(dH)$.

Assume that the Hamiltonian H is given by

$$H(q^i, p_i, S) = \frac{1}{2} g^{ij} p_i p_j + V(q, S) \quad (30)$$

where (g^{ij}) is positive semi-definite (for instance, it is associated to a Riemannian metric on Q). Then, the vector field $\mathcal{E}_H$ describes an isolated simple thermodynamical system with friction satisfying the first and second laws of thermodynamics:

Proposition The integral curves of $\mathcal{E}_H$ satisfies the following properties:

① **First law of Thermodynamics:**

$$\frac{dH}{dt} = 0 \quad (\text{preservation of the total energy});$$

② **Second law of Thermodynamics:**

$$\frac{dS}{dt} = \Delta H \geq 0 \quad (\text{total entropy of an isolated system never decreases}).$$

Proof It is a direct consequence of the above Proposition and $\Delta H = p_i g^{ij} p_j \geq 0$.

If we express the dynamics in terms of the brackets defined in (29) we have that

$$\dot{f} = \{f, H\}_{T^*Q} + \{f, H\}_{\Delta}. \quad (31)$$

Obviously,

$$\{H, H\}_{T^*Q} = \{H, H\}_{\Delta} = 0 \text{ (first law)}$$

and

$$\{S, H\}_{T^*Q} = 0 \text{ and } \{S, H\}_{\Delta} = \Delta H \geq 0 \text{ (second law).}$$

Observe that in Equation (31) both brackets are using the function H as “generator”. This is the reason that typically this formalism is known as *single generator formalism*.

Linearly damped systems

Consider a linearly damped system described by coordinates (q, p, S) , where q represents the position, p the momentum of the particle and S is the entropy of the surrounding thermal bath. We assume that the system is subjected to a viscous friction force, proportional to the minus velocity of the particle. The system is described by the Hamiltonian

$$H(q, p, S) = \frac{p^2}{2m} + V(q) + \gamma S, \quad \gamma > 0.$$

Therefore, the equations of motion for $\mathcal{E}_H = \sharp_\Lambda(dH)$ are:

$$\begin{pmatrix} \dot{q} \\ \dot{p} \\ \dot{S} \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & -p \\ 0 & p & 0 \end{pmatrix} \begin{pmatrix} V'(q) \\ p/m \\ \gamma \end{pmatrix}$$

or

$$\begin{aligned} \dot{q} &= \frac{p}{m} \\ \dot{p} &= -V'(q) - \gamma p \\ \dot{S} &= \frac{p^2}{m} \end{aligned}$$

In the Lagrangian side we obtain the system given by

$$\begin{aligned} m\ddot{q} &= -V'(q) - \gamma m\dot{q} \\ \dot{S} &= m\dot{q}^2. \end{aligned}$$

Observe that in this system the friction force is given by the map $F_{fr} : TQ \rightarrow T^*Q$ given by

$$F_{fr}(q, \dot{q}) = \gamma \dot{q}^i dq^i.$$

Therefore, the equation of entropy production can be rewritten in terms of the friction force as follows

$$T\dot{S} = -\langle F_{fr}(q, \dot{q}), \dot{q} \rangle$$

where $T = \frac{\partial H}{\partial S} = -\frac{\partial L}{\partial S} = \gamma > 0$ represents the temperature of the thermal bath. These equations coincide with the set of equations proposed by Gay-Balmaz and Yoshimura for this particular choice of Lagrangian L and friction force F_{fr} . Observe that, in this particular example where the temperature satisfies $T = \gamma$, the equations are only defined for values $\gamma > 0$ and thus we are only modelling thermodynamical systems with non-zero temperature.

Observe that the two brackets are:

$$\begin{aligned}\{f, g\}_{\Lambda_0} &= \frac{\partial f}{\partial p} \frac{\partial g}{\partial q} - \frac{\partial g}{\partial p} \frac{\partial f}{\partial q} \\ \{f, g\}_{\Delta} &= p \frac{\partial g}{\partial S} \frac{\partial f}{\partial p} - p \frac{\partial f}{\partial S} \frac{\partial g}{\partial p}\end{aligned}$$

In particular

$$\begin{aligned}\{H, g\}_{\Lambda_0} &= \frac{p}{m} \frac{\partial g}{\partial q} - \frac{\partial g}{\partial p} V'(q) \\ \{H, g\}_{\Delta} &= \frac{p^2}{m} \frac{\partial g}{\partial S} - \gamma p \frac{\partial g}{\partial p}\end{aligned}$$

and

$$\mathcal{E}_H(g) = \dot{g} = \{H, g\}_{\Lambda_0} + \{H, g\}_{\Delta}$$

Therefore it is clear that $\{H, H\}_{\Lambda_0} = 0$ and $\{H, H\}_{\Delta} = 0$ (by skew-symmetry) and $\{H, S\}_{\Lambda_0} = 0$ and $\{H, S\}_{\Delta} = \frac{p^2}{m} \geq 0$.

Composed thermodynamical systems without friction

We will present a model for systems composed of at least two subsystems exchanging heat with each other.

Consider two thermodynamic systems indexed by 1 and 2 which may interact through a conducting wall. On each system we have defined the corresponding Hamiltonian:

$$H_i : T^*Q_i \times \mathbb{R} \longrightarrow \mathbb{R}, \quad 1 \leq i \leq 2$$

where (q_i, p_i, S_i) are coordinates on $T^*Q_i \times \mathbb{R}$, $i = 1, 2$, and S_i are the entropies of each subsystem.

Consider the total energy $H : T^*(Q_1 \times Q_2) \times \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

$$H(q_1, p_1, q_2, p_2, S_1, S_2) = H_1(q_1, p_1, S_1) + H_2(q_2, p_2, S_2)$$

and the symplectic 2-form on $T^*(Q_1 \times Q_2) \times \mathbb{R}^2$

$$\Omega_{Q_1 \times Q_2 \times \mathbb{R}^2} = \omega_{Q_1} + \omega_{Q_2} + dS_1 \wedge dS_2$$

Denote by $\Lambda_{Q_1 \times Q_2 \times \mathbb{R}^2}$ the corresponding Poisson tensor.

Assume that both subsystems exchange heat according to Fourier Law:

$$h = k(T_2 - T_1)$$

where T_i is the absolute temperature of subsystem i , $1 \leq i \leq 2$ and k is the coefficient of thermal conductivity. We have that $T_i = \frac{\partial H}{\partial S_i} > 0$.

Consider the function $K : T^*(Q_1 \times Q_2) \times \mathbb{R}^2 \rightarrow \mathbb{R}$

$$K = k \left(\frac{1}{\frac{\partial H}{\partial S_1}} - \frac{1}{\frac{\partial H}{\partial S_2}} \right) = k \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

which will be called *Fourier factor*. Define the Hamiltonian vector field of K by

$$X_K = \Lambda_{Q_1 \times Q_2 \times \mathbb{R}}(\cdot, dK) \quad \text{or alternatively} \quad i_{X_K} \Omega_{Q_1 \times Q_2 \times \mathbb{R}} = dK.$$

Define the two-tensor (with Fourier factor K) denoted by Λ_K , which is associated in the canonical way with the 2-form

$$\Omega_K = \omega_{Q_1} + \omega_{Q_2} + K dS_1 \wedge dS_2$$

Observe that now Ω_K is no longer a symplectic form and so Λ_K is a skew-symmetric Poisson structure. In other words, Λ_K is a skew-symmetric tensor.

The matrix representation of Λ_K is:

$$\begin{pmatrix} 0 & I_{n_1 \times n_1} & 0 & 0 & 0 & 0 \\ -I_{n_1 \times n_1} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & I_{n_2 \times n_2} & 0 & 0 \\ 0 & 0 & -I_{n_2 \times n_2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & k\left(\frac{1}{T_1} - \frac{1}{T_2}\right) \\ 0 & 0 & 0 & 0 & k\left(\frac{1}{T_2} - \frac{1}{T_1}\right) & 0 \end{pmatrix}$$

where $\dim Q_i = n_i$, $i = 1, 2$.

The corresponding evolution vector field $\mathcal{E}_{H,K}$:

$$\mathcal{E}_{H,K} = \sharp_{\Lambda_K}(dH) \quad (32)$$

The integral curves of $\mathcal{E}_{H,K}$ are:

$$\begin{aligned} \dot{q}_1 &= \frac{\partial H}{\partial p_1} & \dot{q}_2 &= \frac{\partial H}{\partial p_2} \\ \dot{p}_1 &= -\frac{\partial H}{\partial q_1} & \dot{p}_2 &= -\frac{\partial H}{\partial q_2} \\ \dot{S}_1 &= k\left(\frac{T_2}{T_1} - 1\right) & \dot{S}_2 &= k\left(\frac{T_1}{T_2} - 1\right). \end{aligned} \quad (33)$$

Observe that the total entropy $S = S_1 + S_2$ satisfies

$$\begin{aligned} \dot{S} &= \mathcal{E}_{H,K}(S_1 + S_2) \\ &= k\left(\frac{T_2}{T_1} - 1\right) + k\left(\frac{T_1}{T_2} - 1\right) \\ &= k \frac{(T_2 - T_1)^2}{T_1 T_2} \geq 0 \end{aligned}$$

Moreover, in absence of external forces, the total energy H is conserved since by skew-symmetry of Λ_K we have that

$$\mathcal{E}_{H,K}(H) = 0 .$$

Thus, we have shown that:

Proposition

Given a Hamiltonian function H , the evolution vector field $\mathcal{E}_{H,K}$ in the skew-symmetric manifold $(T^(Q \times Q \times \mathbb{R}), \Lambda_K)$, which encodes the dynamics of composed thermodynamical systems exchanging heat with each other, satisfies the first and second laws of Thermodynamics.*

Example

The toy model for this case is the two free thermo-particles example, composed by two particles of equal mass such that the system is modelled by the Hamiltonian function

$$H(q_1, p_1, S_1, q_2, p_2, S_2) = \frac{1}{2m}(p_1^2 + p_2^2) + T_1 S_1 + T_2 S_2,$$

where m is the mass of both particles and T_1, T_2 are the temperatures of each particle.

A physical example is the two thermo-spring system, where the system is modelled by the Hamiltonian function

$$H(q_1, p_1, S_1, q_2, p_2, S_2) = \frac{1}{2m}(p_1^2 + p_2^2) + V(q_1, q_2) + T_1 S_1 + T_2 S_2,$$

where V is the potential energy associated to the interplay between both springs, m is the mass of both springs and T_1, T_2 are the temperatures of each spring.

Thank you for your attention!